

Project 1

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Collage: Government Polytechnic Yavatmal

Branch :Computer Engineering

Domain:AI & ML Development

Code:

```
"""Project 1: Student Management System """
```

```
rlno = set()
```

```
database = dict()
```

```
def addstudent():
```

```
    marks = dict()
```

```
    info = dict().fromkeys(["name", "age", "depearnment"], None)
```

```
    no = int(input("Enter the roll no of the student:"))
```

```
    while no in rlno:
```

```
        print("Error , student record already present in the database!")
```

```
        no = int(input("Enter the roll no of the student!"))
```

```
    rlno.add(no)
```

```
    print("Enter the personal information of the student !")
```

```
    info["name"] = input("Enter the full name of the student      :")
```

```
    info["age"] = int(input("Enter the name age of the student      :"))
```

```
    while info["age"] < 0:
```

```
        print("Invalid age:")
```

```
        info["age"] = int(input("Enter the name age of the student  :"))
```

```
    info["depearnment"] = input("Enter the name of your depernment  :")
```

```
    print("Enter the marks of the student in different subjects  :")
```

```
    marks["English"] = float(input("Enter your marks in the english  :"))
```

```
    marks["math1"] = float(input("Enter your marks in the math1  :"))
```

```
    marks["math2"] = float(input("Enter your marks in the math2  :"))
```

```
    marks["Science"] = float(input("Enter your marks in the science :"))
```

```
    marks["Biology"] = float(input("Enter your marks in the Biology :"))
```

```

for i in marks.keys():
    while marks[i] > 100 or marks[i] <0:
        print("Marks must be greater than 0 and smaller than 100")
        marks[i]=float( input(f"Reenter your marks in {i}:"))

```

```

database[no]={"marks":marks,"Information":info}

```

def search():

```

no = int(input("Enter the roll no of the student:"))
if no in rlno:
    print("Information about student is:")
    print("***30)
    print(f"Name      :{database[no]["Information"]["name"]}")
    print(f"age      :{database[no]["Information"]["age"]}")
    print(f"depearnment :{database[no]["Information"]["depearnment"]}")
    print("-"*30)
    print("Marks:")
    print("-"*30)
    for i in database[no]["marks"].keys():
        print("-"*30)
        print(f" | {i}          | {database[no]["marks"][i]} | ")

```

```

else:
    print("Student is not peresent in the Database:")

```

def viewall():

```

for i in database:
    print("Information about student is:")
    print("***30)
    print(f"Name      :{database[i]["Information"]["name"]}")
    print(f"age      :{database[i]["Information"]["age"]}")
    print(f"depearnment :{database[i]["Information"]["depearnment"]}")
    print("-"*30)
    print("Marks:")
    print("-"*30)
    for j in database[i]["marks"].keys():
        print("-"*30)
        print(f" | {i}          | {database[i]["marks"][j]} | ")

```

def update():

```
no = int(input("Enter the roll no of the student to update his record:"))
print("What do you want to update:")
print("1.Information    2.marks")
choice= input("enter the choice:")
if choice=="1" or choice == "information":
    print("1.Name    2.age    3.deperment")
    choice = input("Enter the choice:")
    if choice == "1" or choice=="name":
        database[no]["Information"]["name"]=    input("Enter the new name    :")
    elif choice == "2" or choice=="age":
        database[no]["Information"]["age"]=    input("Enter the new age    :")
    elif choice == "3" or choice=="depearment":
        database[no]["Information"]["depearment"]= input("Enter the new depearment    :")
    else:
        print("Invalid choice!")
        print("-"*30)
```

```
elif choice=="2" or choice == "marks":
    print("1.maths1  2.maths2  3.english  4.science  5.biology")
    choice = input("Enter the choice:")
    if choice=="1" or choice == "maths1":
        database[no]["marks"]["maths1"]=float(input("Enter the new marks:"))
        while database[no]["marks"]["maths1"] >100:
            database[no]["marks"]["maths1"]=float(input("marks must be smaller than 100"))
        print("Sucessfully updated!")
        print("-"*30)
    elif choice=="2" or choice == "maths2":
        database[no]["marks"]["maths2"]=float(input("Enter the new marks:"))
        while database[no]["marks"]["maths2"] >100:
            database[no]["marks"]["maths2"]=float(input("marks must be smaller than 100"))
        print("Sucessfully updated!")
        print("-"*30)
```

```
elif choice=="3" or choice == "english":
    database[no]["marks"]["english"]=float(input("Enter the new marks:"))
    while database[no]["marks"]["english"] >100:
        database[no]["marks"]["english"]=float(input("marks must be smaller than 100"))
    print("Sucessfully updated!")
    print("-"*30)
```

```

elif choice=="4" or choice == "science":
    database[no]["marks"]["science"]=float(input("Enter the new marks:"))
    while database[no]["marks"]["science"] >100:
        database[no]["marks"]["science"]=float(input("marks must be smaller than 100"))
    print("Sucessfully updated!")
    print("-"*30)
elif choice=="5" or choice == "biology":
    database[no]["marks"]["biology"]=float(input("Enter the new marks:"))
    while database[no]["marks"]["biology"] >100:
        database[no]["marks"]["biology"]=float(input("marks must be smaller than 100"))
    print("Sucessfully updated!")
    print("-"*30)
else :
    print("Invalid choice:")
    update()
else :
    print("Invalid choice:")
    update()
return 0

```

```

def delete(no):
    if no in rlno:
        print(f'Removing the student at roll no:{no}')
        print(f'{database.pop(no)} is removed!')
    else:
        print("Student is not present in the database")

```

```

def grad(no):
    sum = 0
    if no in rlno:
        for i in database[no]["marks"].keys():
            sum +=float(database[no]["marks"].get(i))

        return ((sum/500)*100)

```

```

else:
    print("Student is not present in the database!")

```

```
def main():
```

```
while True:
```

```
    print("1. add student    |      *      | 2.view all")
    print("                |   * * *   | ")
    print("3. search student  |   * * * *   | 4.update ")
    print("                |   * *   * *   | ")
    print("5. delete student  | * * * * * * * | 6.calculate gread")
    print("                7.EXIT                ")
```

```
    choice = input("Enter the choice:")
```

```
    print("-"*30)
```

```
    if choice == "7" or choice == "exit":
```

```
        break
```

```
    elif choice == "1" or choice == "add student":
```

```
        addstudent()
```

```
    elif choice == "2" or choice == "view all":
```

```
        viewall()
```

```
    elif choice == "3" or choice == "search student":
```

```
        search()
```

```
    elif choice == "4" or choice == "update":
```

```
        update()
```

```
    elif choice == "5" or choice == "delete student":
```

```
        no = int(input("Enter the roll of the student to delete Record!"))
```

```
        delete(no)
```

```
    elif choice == "6" or choice == "calculate gread":
```

```
        no = int(input("Enter the roll of the student to calculate greade!"))
```

```
        persent = grad(no)
```

```
        if persent > 90 :
```

```
            print( "A+")
```

```
        elif persent > 80 :
```

```
            print("A")
```

```
        elif persent > 70 :
```

```
            print( "B")
```

```
        elif persent > 60 :
```

```
            print( "C")
```

```
        elif persent > 50 :
```

```
            print( "D")
```

```
        elif persent < 40 :
```

```
print( "Fail")
```

```
else:
```

```
print("Invalid Choice!")
```

```
if __name__ == "__main__":  
    main()
```

Output:



PS D:\python programs\Industrial traning> & C:\Users\prita\AppData\Local\Python\pythoncon
ct1.py"

1. add student		*		2.view all
		* * *		
3. search student		* * * *		4.update
		* * * *		
5. delete student		* * * * * *		6.calculate gread
		* * * * * *		
		7.EXIT		

Enter the choice:1

Enter the roll no of the student:80

Enter the personal information of the student !

Enter the full name of the student :pritam santosh tadase

Enter the name age of the student :17

Enter the name of your depernment :computer engineering

Enter the marks of the student in different subjects :

Enter your marks in the english :100

Enter your marks in the math1 :100

Enter your marks in the math2 :100

Enter your marks in the science :100

Enter your marks in the Biology :100

1. add student		*		2.view all
		* * *		
3. search student		* * * *		4.update
		* * * *		
5. delete student		* * * * * *		6.calculate gread
		* * * * * *		
		7.EXIT		

Enter the choice:1

Enter the roll no of the student:56

Enter the personal information of the student !

Enter the full name of the student :vinayakya meshram

```
> ▼ TERMINAL
PS D:\python programs\Industrial traning> & C:\Users\prita\AppData\Local\Python\pythoncon
ct1.py"
Enter the full name of the student      :vinayakya meshram
Enter the name age of the student      :18
Enter the name of your depernment      :aerospace engineering
Enter the marks of the student in different subjects :
Enter your marks in the english       :100
Enter your marks in the math1         :89
Enter your marks in the math2         :90
Enter your marks in the science       :100
Enter your marks in the Biology       :100
1. add student      |          *          | 2.view all
                    |          * * *          |
3. search student  |          * * * *          | 4.update
                    |          * *          * *          |
5. delete student  |          * * * * * * * *          | 6.calculate gread
                    |          7.EXIT          |

Enter the choice:2
-----
Information about student is:
*****
Name      :pritam santosh tadase
age       :17
depearment :computer engineering
-----
Marks:
-----
|80          |100.0|
-----
|80          |100.0|
-----
```

```
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PS D:\python programs\Industrial traning> & C:\Users\prita\AppData\Lo
ct1.py"
|80          |100.0|
-----
|80          |100.0|
-----
|80          |100.0|
-----
|80          |100.0|
-----
|80          |100.0|
Information about student is:
*****
Name       :vinayakya meshram
age        :18
depearment :aerospace engineering
-----
Marks:
-----
-----
|56          |100.0|
-----
|56          |89.0|
-----
|56          |90.0|
-----
|56          |100.0|
-----
|56          |100.0|
-----
1. add student |          *          | 2.view all
                | * * *          |
3. search student | * * * *          | 4.update
```

15

```
PS D:\python programs\Industrial traning> & C:\Users\prita\AppData\Local\Python\python
ct1.py"
```

```

1. add student      |          *          | 2.view all
                    |        * * *       |
3. search student   |      * *   * *     | 4.update
                    |     * *       * *   |
5. delete student   |    * * * * * * * *  | 6.calculate gread
                    |          7.EXIT      |

```

```
Enter the choice:3
```

```
Enter the roll no of the student:80
```

Information about student is:

```
Name      :pritam santosh tadase
```

```
age      :17
```

```
depearment :computer engineering
```

Marks :

English	100.0
---------	-------

```
|math1| 100.0|
```

```
|math2|100.0|
```

```
|Science|100.0|
```

Biology	100.0
---------	-------

```
1. add student      |      *      | 2.view all
```

```
3. search student | * * * * | 4.update
```

```
> ✓ TERMINAL
PS D:\python programs\Industrial traning> & C:\Users\prita\AppData\Local\Python\pythoncore-3.14-ct1.py"

1. add student      |          *          | 2.view all
                   |        * * *        |
3. search student  |      * *   * *      | 4.update
                   |    * *       * *    |
5. delete student  | * * * * * * * * * * | 6.calculate gread
                   |          7.EXIT     |

Enter the choice:4
-----
Enter the roll no of the student to update his record:56
What do you want to update:
1.Information      2.marks
enter the choice:2
1.maths1  2.maths2  3.english  4.science  5.biology
Enter the choice:2
Enter the new marks:100
Sucessfully updated!
-----
1. add student      |          *          | 2.view all
                   |        * * *        |
3. search student  |      * *   * *      | 4.update
                   |    * *       * *    |
5. delete student  | * * * * * * * * * * | 6.calculate gread
                   |          7.EXIT     |

Enter the choice:4
-----
Enter the roll no of the student to update his record:56
What do you want to update:
1.Information      2.marks
enter the choice:1
1.Name          2.age          3.depertment
```

```

TERMINAL

PS D:\python programs\Industrial traning> & C:\Users\prita\AppData\Local\Pyt
ct1.py"

enter the choice:1
1.Name      2.age      3.depertment
Enter the choice:1
Enter the new name      :vinayaka
1. add student |          *          | 2.view all
               | * * *          |
               | * * * * *      | 4.update
               | * *          * * |
               | * * * * * * * * | 6.calculate gread
               | * * * * * * * * |
               | 7.EXIT          |

Enter the choice:5
-----
Enter the roll of the student to delete Record!67
Student is not present in the database
1. add student |          *          | 2.view all
               | * * *          |
               | * * * * *      | 4.update
               | * *          * * |
               | * * * * * * * * | 6.calculate gread
               | * * * * * * * * |
               | 7.EXIT          |

Enter the choice:6
-----
Enter the roll of the student to calculate greade!80
A+
1. add student |          *          | 2.view all
               | * * *          |
               | * * * * *      | 4.update
               | * *          * * |
               | * * * * * * * * | 6.calculate gread
               | * * * * * * * * |
               | 7.EXIT          |

```

```

PS D:\python programs\Industrial traning> & C:\Users\prita\AppData\Local\Python\pythoncore-3.14-64\python.exe "d:/python programs/Industrial traning/proje
ct1.py"
7.EXIT
Enter the choice:6
-----
Enter the roll of the student to calculate greade!80
A+
1. add student      |      *      | 2.view all
                    |    * * *    |
3. search student  |  * * * * *  | 4.update
                    | * *      * * |
5. delete student  | * * * * * * | 6.calculate gread
                    | 7.EXIT
Enter the choice:5
-----
Enter the roll of the student to delete Record!80
Removing the student at roll no:80
{'marks': {'English': 100.0, 'math1': 100.0, 'math2': 100.0, 'Science': 100.0, 'Biology': 100.0}, 'Information': {'name': 'pritam santosh tadase', 'age':
17, 'depearnment': 'computer engineering'}} is removed!
1. add student      |      *      | 2.view all
                    |    * * *    |
3. search student  |  * * * * *  | 4.update
                    | * *      * * |
5. delete student  | * * * * * * | 6.calculate gread
                    | 7.EXIT
Enter the choice:7
-----
PS D:\python programs\Industrial traning>

```